Junaid Saif Khan

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Education

- **Southern Methodist University** — Dallas, Texas, USA
  Doctor of Philosophy in Physics — *June 2023 – Present*
  Cumulative GPA: 4.0 / 4.0
- **The University of Texas at Dallas** — Richardson, Texas, USA
  Master of Science in Physics — *August 2021 – May 2023*
  Cumulative GPA: 3.66 / 4.0
  Graduated 1st in Class
- **Lahore University of Management Sciences** — Lahore, Pakistan
  Master of Science in Physics — *August 2016 – June 2018*
  Cumulative GPA: 3.45 / 4.0
  Ranked 3rd in Class
- **University of the Punjab** — Lahore, Pakistan
  Bachelor of Science in Computational Physics — *October 2012 – June 2016*
  Cumulative GPA: 3.53 / 4.0
  Ranked 3rd out of 45 Students

Research Experience

- **NSF MCgen Research Fellow** — Jan. 2026 – Present
  University of Cincinnati — *Cincinnati, Ohio, USA*
  - **Pre-Fellowship Research & Preparation (Jan. 2026 – Oct. 2026):**
  - Completed foundational Monte Carlo event generator tutorials and currently conducting a targeted literature review to build the project-specific expertise required for model development.
  - Developing machine learning-driven flavor reweighting mechanisms (Shomer/MLHAD) in Pythia string hadronization to enhance heavy-hadron production modeling.
  - Integrating neural network-based reweighting architectures into Monte Carlo event generator workflows to extend modeling capabilities beyond standard pion production.
  - Benchmarking machine learning-based hadronization models against high-energy experimental datasets in collaboration with University of Cincinnati researchers.
  - **Official On-Site Fellowship Term (Nov. 2026 – Apr. 2027):**
  - Transitioning to full-time, on-site research to advance model development and experimental validation.
- **Ph.D. Dissertation Research** — Jun. 2023 – Present
  Southern Methodist University — *Dallas, Texas, USA*
  - **Advisor**: Dr. Fred Olness
  - Studying nonlinear QCD at small Bjorken-x, connecting parton (DGLAP) and dipole (BK) frameworks to investigate gluon saturation.
  - Developed and benchmarked numerical tools (`rcBK`, `bkEval`) to extract dipole cross-sections and structure functions from HERA data.
  - Supporting the Electron-Ion Collider (EIC) program by improving theoretical modeling of small-x deep inelastic scattering.
  - Collaborating with the SURGE (Saturated Glue) Topical Collaboration, associated with Brookhaven National Lab, on global QCD fits.

- **Provost's Summer Research Grant Recipient** May – Aug. 2026
  Office of the Provost, Southern Methodist University *Dallas, Texas, USA*
  - Awarded a competitive university grant dedicated to accelerating doctoral dissertation research and degree completion.
  - Transitioning core machine learning architectures into a scalable, production-ready computational framework for experimental data analysis.
  - Collaborating with experimental groups at Brookhaven National Laboratory (BNL) to embed predictive models into active detector workflows.
  - Integrating advanced theoretical QCD insights into the simulation pipeline through planned residency at the Stony Brook Summer School.
  - Participating in the Boston University AI Workshop to implement state-of-the-art deep learning optimization techniques and finalize datasets for publication.

- **O'Donnell Institute Graduate Research Fellow** Sep. 2025 – May 2026
  O'Donnell Data Science and Research Computing Institute *Dallas, Texas, USA*
  - Awarded fellowship to advance machine learning approaches for nonlinear QCD evolution at the Electron–Ion Collider.
  - Developed machine learning models (MLP and DeepONet) trained on nonlinear QCD solvers, targeting a 100-fold speedup with $\leq$2% error.
  - Integrated surrogate models into global data-fitting pipelines using experimental datasets such as HERA and projected EIC data.
  - Advanced hybrid physics+ML workflows using HPC-generated training data and optimized ML inference.
  - Contributed to the Institute's mission in AI/ML, data science, and high-performance computing through presentations and collaborations.

- **Graduate Researcher – Gravitational Lensing of Binary Objects** Jun. 2022 – May 2023
  The University of Texas at Dallas *Richardson, Texas, USA*
  - **Advisor**: Dr. Lindsay King
  - Simulated gravitational lensing effects on electromagnetic and gravitational wave signals from compact object mergers using general relativity.

- **Young Scientist – Science for Every Seeker** Jun. 2020
  Council for the Lindau Nobel Laureate Meetings *Germany*
  - Selected as Lindau alumnus and young scientist to participate in a 48-hour global Sciathon.
  - Collaborated on science policy challenges during the COVID-19 pandemic.
  - Co-developed the project to promote diversity in STEM; submitted a report and video presentation to the Lindau Council.
  - Designed a prototype for a global mentorship and resource platform for underrepresented researchers.

- **Research Intern – ADM Formulation of General Relativity** May 2020
  Lahore University of Management Sciences *Lahore, Pakistan*
  - Derived the Hamiltonian structure of general relativity using the ADM formalism for canonical quantization.
  - Implemented Dirac's method on symmetry-reduced mini-superspace models in the context of quantum cosmology.

- **Master's Thesis – D-Branes and AdS/CFT** May 2017 – May 2018
  Lahore University of Management Sciences *Lahore, Pakistan*
  - **Advisor**: Dr. B. A. Qureshi
  - Established the AdS/CFT correspondence by comparing correlation functions in Super Yang-Mills theory and supergravity.
  - Explored holographic duality in string theory through D-brane dynamics in type IIB backgrounds.

- **Research Project – Mean Lifetime of Muons** Aug. 2016 – Dec. 2016
  Lahore University of Management Sciences *Lahore, Pakistan*
  - Conducted a cosmic ray experiment to measure muon decay lifetimes from primary atmospheric particles.
  - Operated and calibrated detection equipment including NIM modules, TAC, and PicoScope for timing measurements.
  - Processed data using MCA software and performed decay curve analysis with MATLAB.
  - Achieved experimental muon lifetime within 3% of the theoretical prediction, validating the setup.

- **Research Project – Thermal Diffusive Waves in a Metal** Aug. 2016 – Dec. 2016
  Lahore University of Management Sciences *Lahore, Pakistan*
  - Conducted Fourier analysis of thermal wave propagation in metal using LabVIEW-controlled heat input.
  - Calculated thermal diffusivity and damping coefficients in MATLAB and validated absence of energy transport in diffusive waves.

- **Senior-Year Project – Smeared Potential Model** Jan. 2015 – Jun. 2015
  University of the Punjab *Lahore, Pakistan*
  - **Advisor**: Prof. Bilal Masud
  - Implemented a Coulombic plus linear QCD potential model with smeared hyperfine interaction to study charmonium spectra.
  - Used Crank–Nicolson method to numerically solve the Schrödinger equation.
  - Calculated charmonium meson masses within 1% deviation from experimental data; matched J/ψ mass within 0.016%.

Fellowships, Grants & Honors

About $61,000 in competitive fellowships, grants and travel awards since 2024.

FELLOWSHIPS & RESEARCH GRANTS

- **NSF MCgen Research Fellowship** Nov. 2026 – Present
  University of Cincinnati, Cincinnati, Ohio, USA *$23,500*
  - Awarded a highly competitive NSF-funded external fellowship through the Monte Carlo General Education Network (MCgen) to support advanced training and research in particle and nuclear physics.
  - Selected through a rigorous national review process as one of only four fellows nationwide, collaborating across host sites at the University of Cincinnati and UC Berkeley.

- **Provost's Summer Research Grant** May – Aug. 2026
  Office of the Provost, Southern Methodist University, Dallas, Texas, USA *$5,000*

- **O'Donnell Institute Graduate Student Research Fellowship** Sep. 2025 – May 2026
  O'Donnell Data Science and Research Computing Institute, Dallas, Texas, USA *$24,750*

- **Alpha Chi "In Scholarship and In Service" Grant** Nov. 2025
  Alpha Chi National College Honor Society, Dallas, Texas, USA *$200*
  - Selected as Region II Grant recipient for academic merit and service.

- **Alpha Chi National College Honor Society – Lifetime Membership** Nov. 2024 – Present
  Southern Methodist University *Dallas, Texas, USA*
  - Recognized for academic excellence with a 4.0 CGPA.
  - Honors the top 5% of students across 300+ U.S. chapters.

- **69th Lindau Nobel Laureate Meeting (Physics)** Jun. – Jul. 2019
  Council for the Lindau Nobel Laureate Meetings *Germany*
  - Only master's graduate from Pakistan selected to attend this conference.

- **Department for International Development (DFID) Scholarship** Aug. 2016 – Jun. 2018
  Lahore University of Management Sciences, Lahore, Pakistan *$7,000*
  - Full-tuition scholarship with stipend awarded for master's studies.
- **Merit Scholarship** Sep. 2010 – Apr. 2012
  Punjab Group of Colleges, Lahore, Pakistan *$2,300*
  - Awarded full academic scholarship for intermediate studies based on national merit.

TRAVEL & CONFERENCE GRANTS

- **NSF IAIFI 2026 Travel Grant** Aug. 2026
  Summer School at the University of Massachusetts, Boston, Massachusetts, USA *$1,500*
- **EIC Theory Institute at Brookhaven National Lab Travel Grant** Jun. 2026
  Brookhaven National Lab, Upton, New York, USA *$500*
- **Center for Frontiers in Nuclear Science (CFNS) Travel Grant** Jun. 2026
  Summer School at Stony Brook University, Stony Brook, New York, USA *$1,000*
- **Catalyzing Advocacy in Science and Engineering (CASE) Travel Grant** Apr. 2026
  Moody School of Graduate and Advanced Studies and Dedman College of Humanities and Sciences *$1,500*
  - Awarded competitive funding from the Moody School of Graduate and Advanced Studies and Dedman College to support travel and participation in the 2026 AAAS CASE Workshop in Washington, D.C.
- **Artificial Intelligence for the Electron Ion Collider (2025) Travel Grant** Oct. 2025
  Conference at MIT, Cambridge, Massachusetts, USA *$950*
- **Hampton University Graduate Studies Program (HUGS) Travel Grant** May – Jun. 2025
  Summer School, Jefferson Lab, Newport News, Virginia, USA *$980*
- **Heavy-Flavor Theory (HEFTY) Travel Grant** Jun. 2024
  Summer School, Santa Fe, New Mexico, USA *$1,100*

Software & Computational Experience

- **Programming Languages:** Python, C++, C#, Mathematica, MATLAB, Bash/Shell, LaTeX.
- **Machine Learning & Data Science:** PyTorch, Keras, Scikit-learn, NumPy, Pandas; trained surrogate ML models (DeepONet, MLP) for nonlinear QCD solvers.
- **Monte Carlo & Numerical Methods:** Developed and benchmarked stochastic solvers (rcBK, bkEval) for nonlinear QCD evolution; experience with Monte Carlo integration, uncertainty quantification, and data fitting using **xFitter**.
- **High-Performance Computing (HPC):** Parallelized ML training and numerical simulations on multi-node clusters (SMU ManeFrame II); optimized inference for large-scale datasets.
- **Scientific & Visualization Tools:** ROOT, LabVIEW, SciPy, Matplotlib, Adobe Illustrator; experience with simulation output visualization and scientific plotting.
- **Version Control & Workflow:** Git, GitHub, and collaborative development pipelines for code management and reproducible research.

Publications

- **Khan, J. S.**, Lustberg, R. L., Olness, F., Risse, P., Schenke, B., and Stevenson, B. "*Probing Dense Nuclear Matter at Small-x: A workflow for a global analysis framework*", 33rd International Workshop on Deep Inelastic Scattering and Related Subjects (DIS2026), Bologna, Italy, arXiv:2609.00230 [nucl-th].
- **Khan, J. S.**, et al. "*Machine Learning Surrogates for Nonlinear QCD Evolution in Dense Nuclear Matter*", *in preparation* (2026).
- Asad, F., **Khan, J. S.**, Tariq, M., et al. "*Federated Proximal Optimization for Privacy-Preserving Heart Disease Prediction: A Controlled Simulation Study on Non-IID Clinical Data*", arXiv:2601.17183 [cs.LG], 2026.

Presentations

CONTRIBUTED PRESENTATIONS

- **Bridging QCD Theory and ML for Next-Gen Nuclear Physics** Aug. 2026
  University of Massachusetts Boston *Boston, Massachusetts, USA*
  - Shared methodologies for embedding AI frameworks into traditional physics pipelines, paving the way for more efficient calculations in upcoming experimental programs.

- **Small-x Dynamics at the EIC via Machine Learning** Jun. 2026
  EIC Theory Institute, Brookhaven National Laboratory *Upton, New York, USA*
  - Invited talk

- **Nonlinear Parton Dynamics and the Role of ML Surrogate Models** Jun. 2026
  Center for Frontiers in Nuclear Science (CFNS) Summer School *Stony Brook, New York, USA*
  - Presented research on fast neural network models to simulate dense gluons in high-energy particle collisions.

- **Applications of AI/ML in Nuclear and Particle Physics** Jan. 2026
  HEFTY Collaboration Meeting *Dallas, Texas, USA*
  - Delivered a research talk highlighting LLM-based knowledge systems for large experiments, ML-driven particle identification, and fast surrogate models for low-x QCD global analyses.

- **Nonlinear QCD Evolution with Machine Learning for the EIC** Oct. 2025
  Texas, Arkansas, Oklahoma Section of the APS, Texas Christian University *Fort Worth, Texas, USA*
  - Delivered a talk on applying machine learning to nonlinear QCD evolution for the Electron-Ion Collider.

- **Unraveling QCD Puzzles via the Electron–Ion Collider (EIC)** Oct. 2025
  Texas, Arkansas, Oklahoma Section of the APS, Texas Christian University *Fort Worth, Texas, USA*
  - Presented a poster introducing the EIC and its role in unraveling QCD puzzles.

- **Nonlinear QCD at Small-x: Parton Model vs Dipole Model** May – Jun. 2025
  HUGS Summer School, Jefferson Lab *Newport News, Virginia, USA*
  - Awarded 2nd Prize in the HUGS poster session.

- **Exploring Gluon Dynamics in the Small-x Regime of QCD** Apr. 2025
  SMU Research and Innovation Week *Dallas, Texas, USA*
  - Presented poster highlighting theoretical work on gluon saturation in QCD.

- **QCD in the Non-Perturbative Region: Gluons in Small-x Regime** Jun. 2024
  HEFTY Summer School *Santa Fe, New Mexico, USA*
  - Delivered a contributed talk on small-x QCD and gluon saturation, in collaboration with BNL.

- **Phenomenology of String Cosmology** Aug. 2023
  Southern Methodist University *Dallas, Texas, USA*
  - Course project on inflationary dynamics in string-inspired cosmology.

- **Gravitational Lensing of Compact Binary Objects and their Host Galaxies** Oct. 2022
  Theoretical Astroparticle and Cosmology Symposium in Texas, SMU *Dallas, Texas, USA*
  - Explained lensing models for compact binaries as part of a graduate course presentation.

- **Emergence and Correspondence of Holography and AdS/CFT** Jan. 2020
  Lahore University of Management Sciences (LUMS) *Lahore, Pakistan*
  - Poster presentation communicating duality principles studied during string theory research.

- **Harnessing Cosmic Ray Muons** Dec. 2016
  Fourth Regional Lab Immersion Program, LUMS *Lahore, Pakistan*
  - Presented muon detection methods using scintillator-based systems.

INVITED & OUTREACH TALKS

- **Small-x Dynamics at the EIC via Machine Learning** Jun. 2026
  EIC Theory Institute, Brookhaven National Laboratory *Upton, New York, USA*
  - Invited talk on utilizing neural network surrogates to emulate nonlinear small-x QCD dynamics and accelerate global analysis workflows.

- **Absurdity of Infinity in Physics** Jul. 2024
  QuarkNet Annual Workshop, SMU *Dallas, Texas, USA*
  - Delivered an interactive talk to high school physics teachers across Texas, exploring the conceptual challenges of infinity in modern theoretical physics.

- **Introduction to String Theory & Dualities** Jun. 2020
  University of Management and Technology (UMT) *Lahore, Pakistan*
  - Invited departmental talk on core dualities in modern string theory.

- **Road to String Theory & Holography** Jun. 2020
  Online Public Seminar *Virtual*
  - Delivered an outreach lecture introducing the ideas behind string theory and the holographic principle.

- **Image of a Black Hole: Faith or Science?** Sep. 2019
  National University of Computer & Emerging Sciences (FAST) *Lahore, Pakistan*
  - Public talk on the scientific impact of the Event Horizon Telescope and M87.

Conferences, Workshops & Seminars Attended

- **NSF IAIFI Summer School** Aug. 2026
  University of Massachusetts Boston *Boston, Massachusetts, USA*
  - Selected participant in an interdisciplinary summer school on AI and fundamental physics, featuring lectures, hands-on coding tutorials, collaborative hackathon projects, and global research networking.

- **2026 CFNS Summer School on the Physics of the Electron-Ion Collider (EIC)** Jun. 2026
  Stony Brook University *New York, USA*
  - Intensive two-week program on EIC phenomenology, small-x QCD, and gluon saturation physics.

- **Catalyzing Advocacy in Science and Engineering (CASE) Workshop** Apr. 2026
  American Association for the Advancement of Science (AAAS) *Washington, D.C.*
  - Selected through a competitive university-wide selection process to participate in intensive training on the federal budget process, science communication, and the intersection of U.S. science and technology policy.
  - Represented SMU in meetings with Congressional staff to advocate for federal research funding and the importance of national laboratory infrastructure for the future STEM workforce.

- **Artificial Intelligence for the Electron Ion Collider (2025)** Oct. 2025
  Massachusetts Institute of Technology *Cambridge, Massachusetts, USA*
  - Participated in sessions on AI applications in detector design, data reconstruction, and EIC simulations.
  - Gained insight into AI-driven approaches for event reconstruction and modeling in collider physics.

- **The 1st International Workshop on TeV Muon-Ion Colliders** Dec. 2023
  Rice University *Houston, Texas, USA*
  - Attended sessions on physics, accelerator design, and experimental challenges for muon-ion colliders.

- Complemented EIC research by deepening insight into TeV-scale scattering.

- **Fall 2023 Joint Meeting of the Texas Section of the APS** Oct. 2023
  Angelo State University *San Angelo, Texas, USA*
  - Attended the regional APS meeting, engaging with researchers, faculty, and students throughout Texas.

- **Theoretical Astroparticle and Cosmology Symposium in Texas** Oct. 2022
  Southern Methodist University *Dallas, Texas, USA*
  - Attended sessions on cosmology, dark matter, and high-energy theory at the particle–astrophysics interface.

- **String Phenomenology 2020** Jun. 2020
  Northeastern University *Virtual (USA)*
  - Engaged in virtual talks on string compactifications, swampland conjectures, and AI in theoretical physics.

- **The Salam Sessions** Sep. – Dec. 2019
  Lahore University of Management Sciences *Lahore, Pakistan*
  - Chapter 1: Discourse on "*The Effectiveness of Mathematics in the Physical Sciences*" (Sep. 2019).
  - Chapter 2: "*Spectral Theory of Operators and Quantum Mechanics*" (Oct. 2019).
  - Chapter 3: Talk on Eugene Wigner's "*The Unreasonable Effectiveness of Mathematics*" (Dec. 2019).

- **Information, Black Holes and Quantum Theory** Mar. 2018
  Abdus Salam School of Mathematical Sciences (ASSMS) *Lahore, Pakistan*
  - Workshop exploring GR and QFT in curved spacetime, Hawking radiation, and holography.

- **Representations of Lorentz Group and Non-Abelian Gauge Theories** Feb. 2017
  Symposium on Theoretical Physics, ASSMS *Lahore, Pakistan*

- **The Higgs Boson and Salam's Model of Particle Physics** Dec. 2016
  Seminar at Lahore University of Management Sciences *Lahore, Pakistan*

- **8th International Meetings on Particles and Fields** Apr. 2016
  COMSATS Institute of Information Technology *Lahore, Pakistan*

Teaching Experience

Created and organized course materials, instructed **850+ undergraduate** and **8 graduate students** across **10 laboratory** and **13 lecture courses**, led discussions, and evaluated performance with a **99% success rate**.

- **Southern Methodist University** Aug. 2023 – Aug. 2025
  Graduate Teaching Assistant, Department of Physics *Dallas, Texas, USA*
  - **Mechanics Laboratory**
  - **Electricity and Magnetism Laboratory**
  - **Introductory Mechanics**

- **The University of Texas at Dallas** Aug. 2021 – May 2023
  Graduate Teaching Assistant, Department of Physics *Richardson, Texas, USA*
  - **Mechanics, College Physics I**
  - **Physics Laboratory II**

- **Lahore University of Management Sciences (LUMS)** Jan. 2017 – May 2021
  Teaching Assistant, Physics Department *Lahore, Pakistan*
  - **Modern Physics**
  - **Mechanics**
  - **Introduction to Quantum Field Theory**

- **General Relativity**
- **Electricity and Magnetism**

- **National University of Computer and Emerging Sciences** Aug. 2018 – Feb. 2021
  Visiting Lecturer, Department of Sciences & Humanities *Lahore, Pakistan*
  - **Physics for Engineers**
  - **Applied Physics**

Professional Affiliations

- **Brookhaven National Lab** Dec. 2023 – Present
  SURGE Collaboration Member *Upton, New York, USA*
- **Jefferson Lab** May – Jun. 2025
  Member *Newport News, Virginia, USA*
- **Alpha Chi National College Honor Society** Nov. 2024 – Present
  Lifetime Member *USA*
- **American Physical Society** Oct. 2023 – Present
  Member *USA*

Volunteering & Community Services

- **SMU Research and Innovation Week** Apr. 2025
  Southern Methodist University *Dallas, Texas, USA*
  - Poster Judge for undergraduate research projects.
- **Dallas Regional Science and Engineering Fair (DRSEF)** Feb. 2025
  Texas Education Agency Region 10 Annual Science Fair *Dallas, Texas, USA*
  - Judged STEM poster presentations from high school students.
- **Total Solar Eclipse Event** Apr. 2024
  Southern Methodist University *Dallas, Texas, USA*
  - Local organizing committee member.
- **Summer Platform for Undergraduate Research (SPUR)** Jul. 2022
  The University of Texas at Dallas *Richardson, Texas, USA*
  - Poster Judge; reviewed over 10 undergraduate student projects in STEM disciplines.
- **CERN LHC Interactive Tunnel** Jan. 2020 & Oct. 2019
  Children's Literature Festival & Lahore Science Festival *Lahore, Pakistan*
  - Led outreach as Head Organizer in collaboration with CERN and Khwarizmi Science Society.
  - Engaged public with interactive demonstrations promoting science outreach.
- **Lahore Science Festival** Jan. 2017
  Khwarizmi Science Society *Lahore, Pakistan*
  - Demonstrated physics exhibits including Newton's Cradle, Lichtenberg pattern, and Van de Graaff generator.

Leadership Experience

- **LUMS Students Mathematics Society** Sep. 2017 – Jun. 2018
  Member, Lahore University of Management Sciences *Lahore, Pakistan*
  - Organized campus-wide STEM outreach events such as math scavenger hunts and quiz competitions.
  - Arranged invited talks featuring scientists from across Pakistan.